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Generative AI and the model design space

Where does DeepNLME live, and what can we borrow from machine learning?

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WHERE WE ARE

So far we have collected a box of seemingly separate tools

YESTERDAY AND THIS MORNING

- **ODEs** — mechanism we write down by hand
- **UDEs / NODEs** — let a neural net learn part (or all) of the dynamics
- **Random effects** — let each subject differ

Each felt like its own trick, with its own fitting story.

THE CLAIM OF THIS SESSION

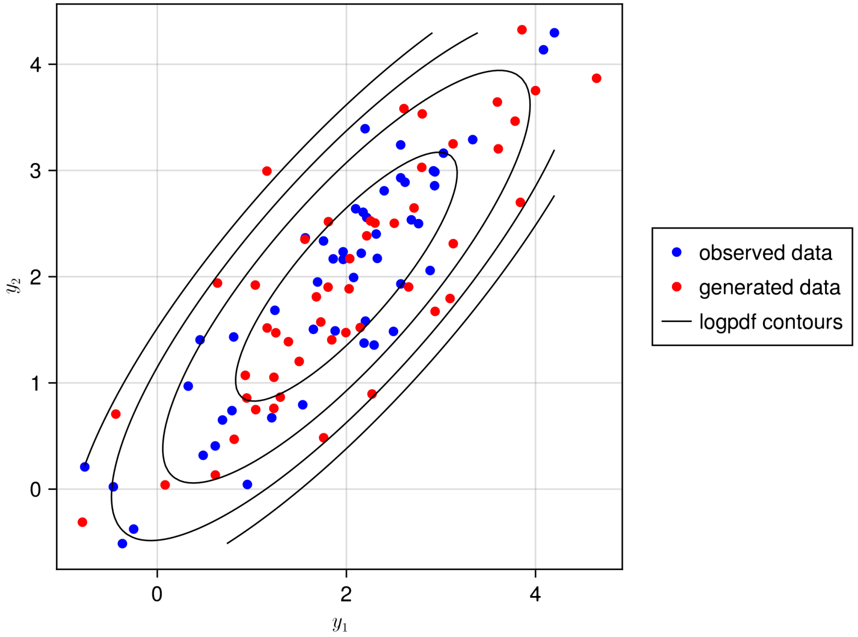
They are all **points in a single design space**.

And the most powerful moves through that space are exactly the ideas that power **generative AI**.

ONE MAP

DeepNLME is not a new gadget — it is one **coordinate** on a map that also contains VAEs and Latent Neural ODEs.

Match the data's distribution — in any space



…even when the space is **faces** — these people don't exist.

Generated samples fall in the **same distribution** as the real data.

A DISTRIBUTION OVER A VERY DIFFERENT SPACE

A generative model learns the **distribution** of the data — scalars, time courses, or images of faces — and draws new samples from it.

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A generative model invents the unobserved part

THE CATCH

Data is a mix of

- **observed** quantities — pixels, tokens, concentrations
- **unobserved** quantities — the face, the meaning, the patient

A model has to invent the unobserved part before it can generate the rest.

THE LATENT-VARIABLE RECIPE

$$y = f(z) + \varepsilon$$
$$z \sim \mathcal{N}(0, I)$$

Draw an unobserved **latent** z, push it through a generator f, add noise. Train f so generated y matches real data in distribution.

THE REFRAME

An NLME model is a conditional generative model

GENERATIVE MODEL

$$\boldsymbol{y} = f(\boldsymbol{z}) + \varepsilon$$
$$\boldsymbol{z} \sim \mathcal{N}(0, I)$$

latent $\boldsymbol{z}$ — “what we can’t see”

NLME MODEL

$$\mathrm{dv} = f_{\theta(\boldsymbol{\eta},\boldsymbol{x})} + \varepsilon$$
$$\boldsymbol{\eta} \sim \mathcal{N}(0, \Omega)$$

random effects $\boldsymbol{\eta}$ — “how this subject differs”

SAME SHAPE

The **random effects are the latent variables**. The structural model f_θ is the generator. We just **condition** on covariates and dose x.

NLME and VAEs maximise the same objective

NLME

$$p(\boldsymbol{y} \mid \boldsymbol{x}) = \int p(\boldsymbol{y} \mid \boldsymbol{\eta}, \boldsymbol{x})\, p(\boldsymbol{\eta})\, d\boldsymbol{\eta}$$

marginalise out the random effects $\boldsymbol{\eta}$

VARIATIONAL AUTO-ENCODER

$$p(\boldsymbol{x}) = \int p(\boldsymbol{x} \mid \boldsymbol{z})\, p(\boldsymbol{z})\, d\boldsymbol{z}$$

marginalise out the latent variables $\boldsymbol{z}$

THE SAME TARGET

Both go after the **marginal likelihood** — integrating over an unobserved latent. Neither reaches it exactly: NLME approximates the integral; the VAE maximises a **lower bound** on it (the ELBO). The latent variables *are* the random effects: $\boldsymbol{z} \equiv \boldsymbol{\eta}$.

WHERE METHODS DIVERGE

That integral is the whole problem

$$p(\boldsymbol{y} \mid \boldsymbol{x}) = \int \underbrace{p(\boldsymbol{y} \mid \boldsymbol{\eta}, \boldsymbol{x})}_{\text{generator / structural model}} \underbrace{p(\boldsymbol{\eta})}_{\text{prior on latents}} \, d\boldsymbol{\eta}$$

WHY IT'S HARD

We never observe η. To score a model we must integrate it out — and that integral has no closed form for a nonlinear f_θ.

THE KEY INSIGHT

Every method below is just a **different way to approximate this one integral**. That choice is what separates NLME, VAEs and Latent NODEs — not the equation.

Two ways to pin down the latent variable

A POSTERIOR PER SUBJECT — NLME

Each subject keeps **its own** posterior, by Bayes' rule:

$$p(\boldsymbol{\eta}_i \mid \boldsymbol{y}_i, \boldsymbol{x}_i) \propto p(\boldsymbol{y}_i \mid \boldsymbol{\eta}_i, \boldsymbol{x}_i)\, p(\boldsymbol{\eta}_i)$$

Represented **per subject** — by FOCE, Laplace, EM/SAEM or MCMC. The method varies; the **per-patient representation** does not.

AMORTISED INFERENCE — VAE

One shared **encoder** outputs the **parameters** of each subject's posterior:

$$q\big(\boldsymbol{\eta}_i\,;\lambda_\varphi(\boldsymbol{y}_i, \boldsymbol{x}_i)\big)$$

The parameters λ are a **deterministic function of the data**; the family of q is free — Gaussian, flow, mixture. One model for all subjects, trained once (ELBO). **No per-subject fit.**

- The shift: from **representing** each patient's posterior individually → **predicting** it with one shared model

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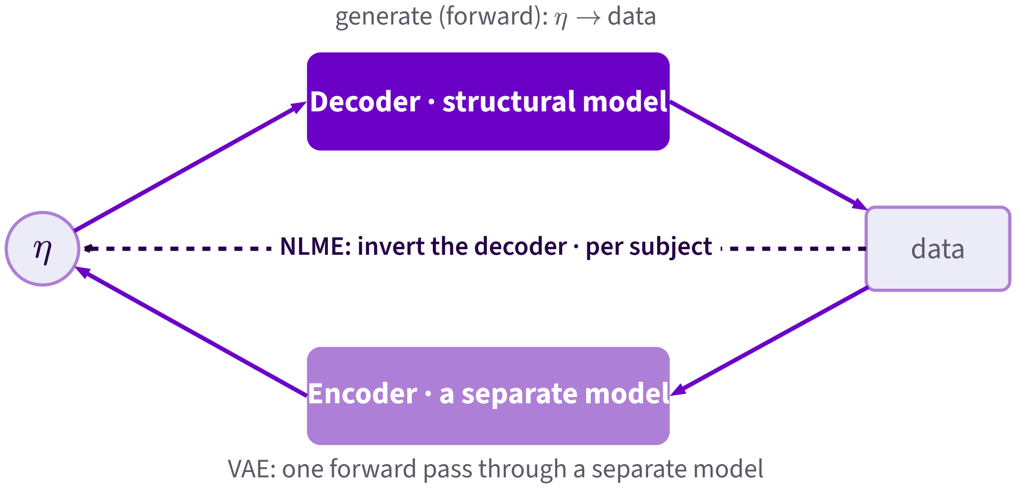
The encoder is the inverse of the decoder



NLME — INVERT THE DECODER

Solve the decoder's **inverse for each subject** — no extra model. **Less bias** (it really solves the inverse), but a **per-subject** optimisation.

VAE — A SEPARATE ENCODER

Train **one** model to do the inverse in a forward pass. **Scales** to very large problems, at the cost of **more bias** (the amortisation gap).

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How much of the generator is mechanism vs. learned?

MECHANISTIC	HYBRID	DATA-DRIVEN
ODE every term written by hand from pharmacology	**UDE** keep the structure you trust, let a neural net learn the rest	**Neural ODE** the whole vector field is a neural network

YOU ALREADY MOVED ALONG THIS AXIS

This morning’s UDE exercise **is** a step from mechanism toward data-driven — one move along this axis.

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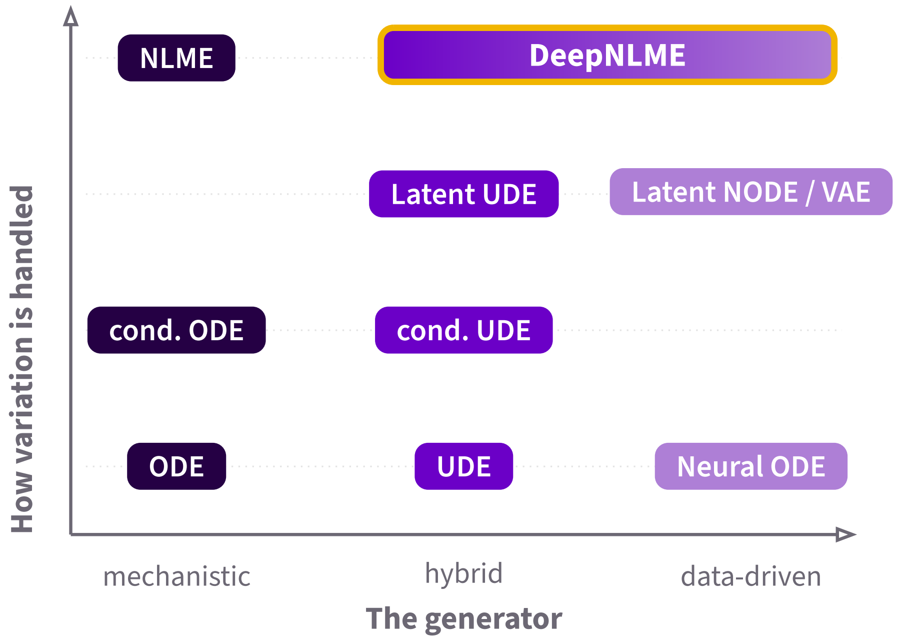
Two axes, one map



Up = richer handling of **between-subject variability**: none → conditional → amortised latents → per-individual random effects.

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READING THE MAP

DeepNLME's neighbours are one move away

DeepNLME = **per-subject random effects** with a generator you can dial from **hybrid** all the way to **fully neural**. From there:

1. **Add full mechanism** (drop the neural net) → plain **NLME** (one move left).
2. **Amortise the inference** → a **Latent UDE**, then a **Latent NODE / VAE** (move down).

"How much mechanism" is a **dial within DeepNLME**, not a jump to a different model.

WHY THIS IS THE SWEET SPOT

DeepNLME keeps the **mechanism you trust** and the **per-subject rigour** of NLME, while borrowing the **flexible function approximation** of deep generative models.

It sits where pharmacology and machine learning overlap — deliberately, not by accident.

THE AMORTISED COUSIN

A Latent Neural ODE is a VAE over trajectories

THE PIPELINE

1. **Encoder** reads a subject's time course → a latent z_0.
2. A **neural ODE** evolves z_0 forward in time.
3. **Decoder** maps the latent trajectory to observations.

Trained exactly like a VAE: an amortised encoder maximising the **ELBO** — a **lower bound** on the marginal likelihood.

SAME MAP, DIFFERENT COORDINATE

Latent NODE = **data-driven generator** + **amortised latents**.

Add mechanism and swap amortised inference for a per-subject posterior and you have walked back to **DeepNLME**.

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SO WHAT

Borrowing from ML = moving around this map

Once you see NLME as a latent-variable generative model, a whole toolbox opens up:

- amortise inference when you have many subjects
- let neural nets fill in unknown mechanism
- **couple** the random-effect latent space to **other** latent-variable models

NEXT: COUPLING LATENT SPACES

If a text or image model also has a latent space, and our NLME has one, we can **link them** — pour rich covariates into the random effects.

That is exactly the **embeddings** exercise, and the **ModelJoin** idea, coming up next.

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TAKEAWAYS

One space, many neighbours

1. An **NLME model is a conditional generative model** — random effects are its latent variables.
2. **NLME and VAEs share the same target — the marginal likelihood**; they differ in how they approximate it (NLME the integral, the VAE a lower bound — the ELBO).
3. **Two axes** organise the field: mechanism ↔ data-driven, and how individual variation is inferred.
4. **DeepNLME** is a dial, not a point — per-subject random effects with a generator ranging from NLME (mechanistic) to fully neural, one move from the amortised Latent NODE / VAE.
5. Moving around the map = **borrowing ML ideas for pharmacometrics**, which is what the rest of the day is about.

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